

ANMOL AGARWAL

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Software Engineer with experience building and maintaining production-grade data and ML pipelines at a large consumer marketplace. Strong background in backend systems, data reliability, and analytics platforms. Comfortable working across Java, Python, and JavaScript with solid foundations in DSA and system design.

WORK EXPERIENCE

Aug'23 – Nov'25 | Personal Goal Pursuit – Civil Services Preparation (UPSC)

Jun'22 – July'23 | Urban Company, Gurgaon | Software Engineer

- **Core Data Engineering & Pipeline Platforms**
 - Designed and implemented **config-driven ETL data and ML pipelines** supporting **batch processing**, enabling **standardized, schema-driven onboarding of data science models**, reducing onboarding turnaround by **~60–70%**.
 - Developed **robust event ingestion and configuration platforms** serving multiple internal consumers with self-serve dashboards and versioned APIs, deprecating legacy flows and reducing **~15 weekly platform support tickets**.
- **Reliability, Backfill & Quality Systems**
 - Identified and **resolved race conditions and idempotency issues** in **data backfilling workflows**, improving **data quality, reliability, and correctness** across dependent downstream layers.
 - Built observability into pipeline runs through **structured logging, monitoring hooks, and alert integrations**, improving operational insight and reducing incident response time.
- **Deployment & Performance Automation**
 - Enabled **CI/CD pipelines** for automated deployments of production **data and ML services**, reducing manual release steps by **~50%** and improving deployment **reliability** and **SLA adherence**.
 - Optimized **Apache Superset** analytics workloads by tuning refresh strategies and queries, increasing cross-team **adoption** while **reducing** unnecessary **compute costs** by **~30%**.

Jan'22 – Apr'22 | Urban Company, Remote | Software Developer Intern

- Migrated a **core backend service** from **monolithic architecture to microservices**, reducing deployment coupling and improving fault isolation across **3–4 independent components**.
- Implemented **audit logging and change-tracking** capturing old/new values, user IDs, and timestamps, covering **100% of critical write operations** and improving system **traceability** and **security**.
- Built **SQL-based alerting on key business and system metrics**, enabling early anomaly detection and reducing manual monitoring effort by **~50%**.

Feb'21 – Jun'21 | Algoji, Remote | Quant Developer Intern

- Developed and deployed **algorithmic trading strategies** (Short Straddle, Fibonacci-based) using Python, xlwings, and schedulers, integrated with broker APIs for **100% automated order execution** across multiple live strategies.
- Optimized strategy execution by analyzing **performance metrics and trader feedback**, reducing order placement **latency by ~30–40%** and improving overall execution efficiency.
- Improved robustness of trading systems by implementing **structured logging, exception handling, and fail-safes**, achieving **99%+ execution uptime** in backtests, and **mentored 15+ interns** in strategy design and optimization.

ACADEMIC PROJECTS

2022: Pairs Trading Using Machine Learning

- Developed a **pairs trading** strategy on **1027 NSE securities** using **PCA, DBSCAN** and used **cointegration test** to find the pairs.
- Identified **48 profitable pairs** out-perform the benchmark with maximum profit of **467%** on single pair **over 5 years (2015-2019)**.

2021: Generation Of Malware And Its Detection

- Implemented static malware detection using **PE32 header features**, extracting and **analyzing 50+ executable attributes** for feature engineering.
- Validated models on self-generated malware samples, **achieving >90% classification accuracy** under controlled experimental conditions.

TECHNICAL SKILLS

- **Languages:** Java | Python | JavaScript | SQL
- **Data & Backend:** Data Pipelines | ETL | Backfilling | Batch & Streaming Concepts | REST APIs | CI/CD
- **Core CS Concepts:** Data Structures & Algorithms | Object-Oriented Programming | Operating Systems | Database Management | System Design
- **Analytics & Tools:** Apache Superset | Pandas | NumPy
- **Version Control & Collaboration:** Git | GitHub | Bitbucket | Jira | Confluence

ACHIEVEMENTS

- **Cleared UPSC Civil Services Preliminary Examination twice**, ranking in the **top ~1–1.4%** among **~1M+ candidates** nationwide.
- **CFA Level I** qualified, demonstrating strong foundations in **finance, economics, and analytical reasoning**.
- **Published Research Paper** - based on **comparative study on pairs trading** with different machine learning models. Results shows that we achieved accuracy of **95% pairs to be profitably** traded in one of the strategies.
- Secured **Top 1%** in **JEE Main (2018)** among **~1.2M candidates**.
- Achieved **City Rank 1** in **SOF International Mathematics Olympiad (2013)**.

EDUCATION

- **2022** : B.Tech. (Computer Engineering) **Delhi Technological University**, Delhi – **CGPA : 8.60**
- **2018** : Class 12th from Lal Bahadur Shastri Sr. Sec. School, Delhi – **Score : 92.2%**
- **2016** : Class 10th from St. Don Bosco Sschool, Lakhimpur – **CGPA : 9.40**